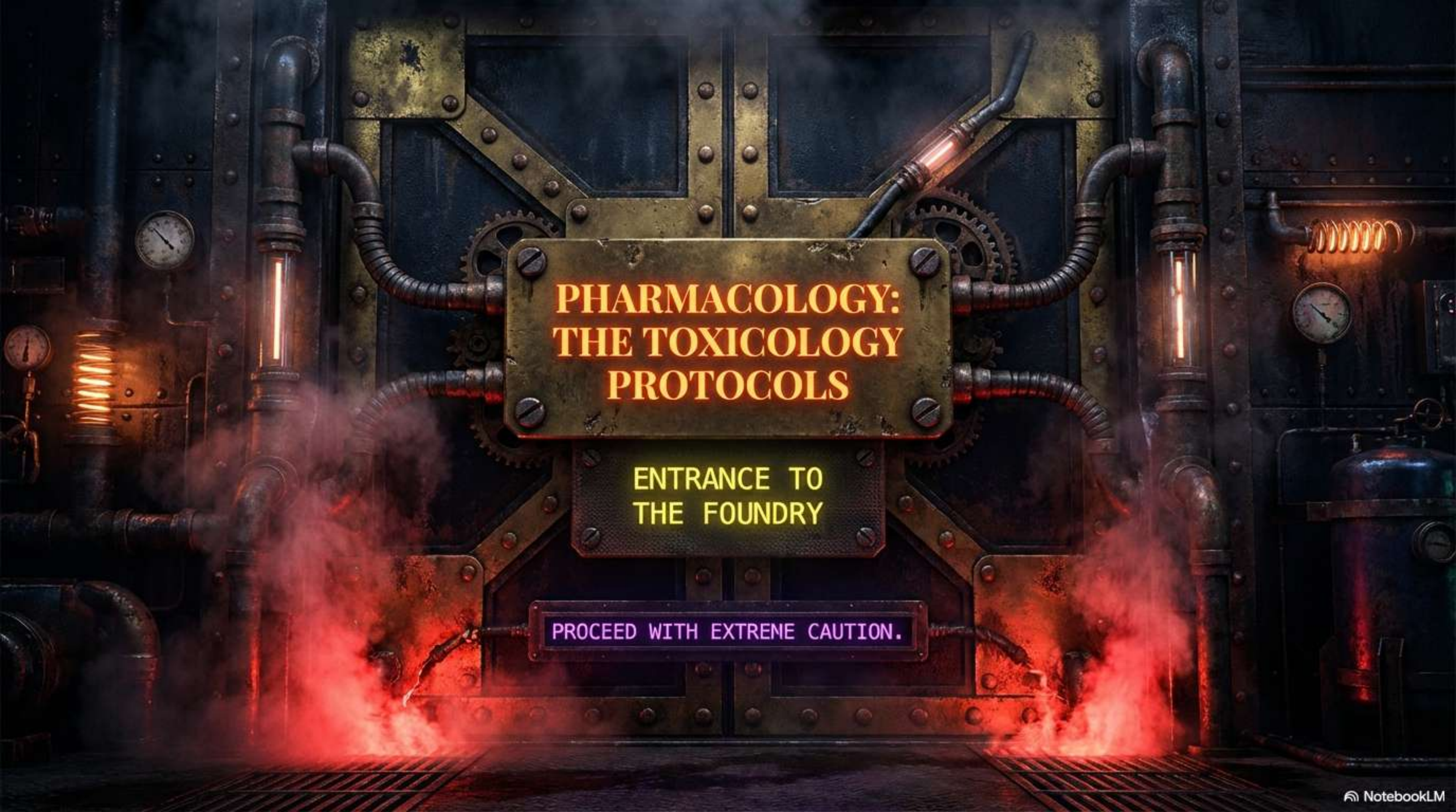
PHARMACOLOGY:
THE TOXICOLOGY
PROTOCOLS
ENTRANCE TO
THE FOUNDRY
PROCEED WITH EXTREME CAUTION.

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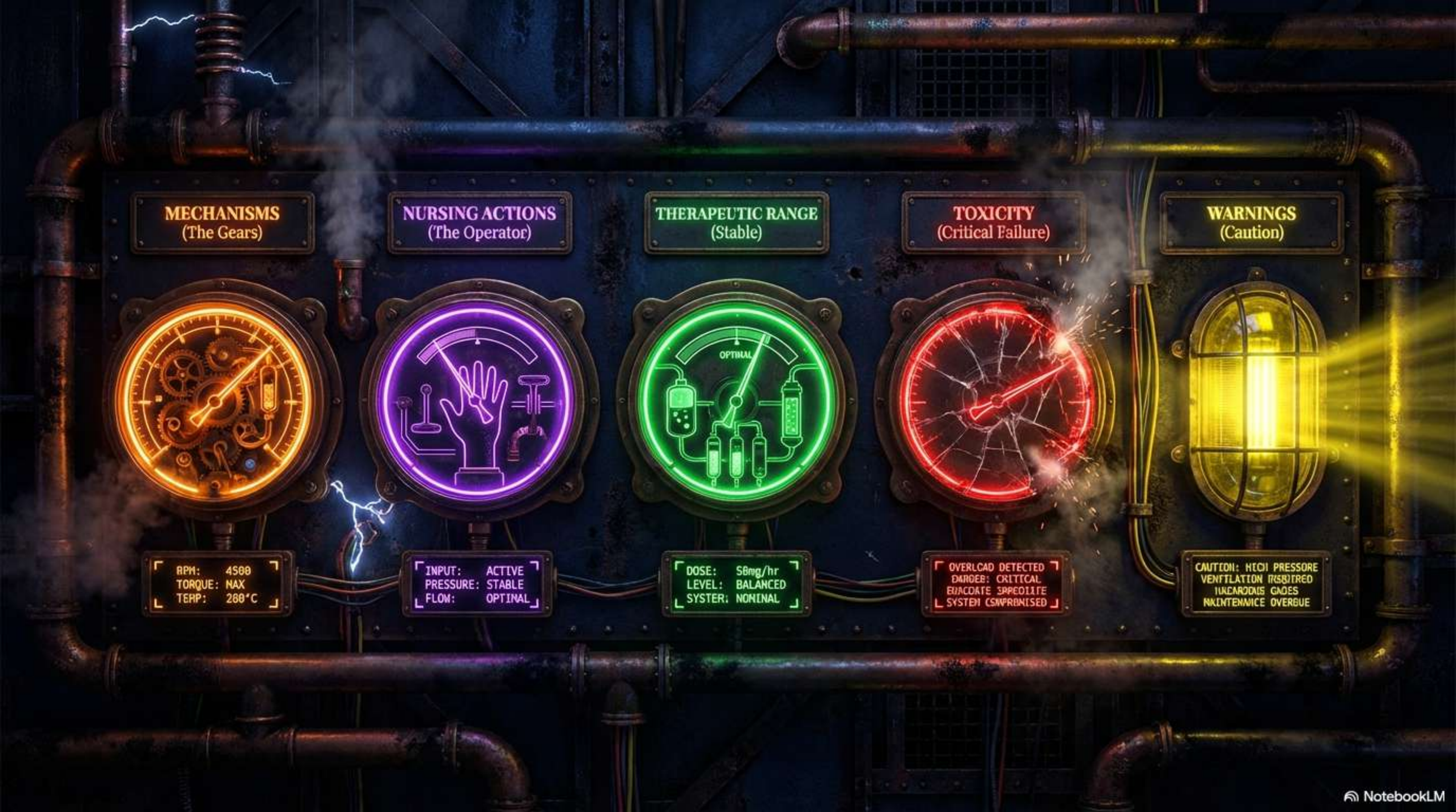
MECHANISMS
(The Gears)
NURSING ACTIONS
(The Operator)
THERAPEUTIC RANGE
(Stable)
TOXICITY
(Critical Failure)
WARNINGS
(Caution)
OPTIMAL
RPM: 4500
TORQUE: MAX
TEMP: 280°C
INPUT: ACTIVE
PRESSURE: STABLE
FLOW: OPTIMAL
DOSE: 50mg/hr
LEVEL: BALANCED
SYSTEM: NOMINAL
OVERLOAD DETECTED
DAMAGE: CRITICAL
EVACUATE IMMEDIATE
SYSTEM COMPROMISED
CAUTION: HIGH PRESSURE
VENTILATION REQUIRED
HAZARDOUS GASES
MAINTENANCE OVERDUE

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Indications: Serious, Life-Threatening, Resistant, Gram-Negative Infections.
AMINOGLYCOSIDES: THE BIG GUNS
A MEAN OLD MYCIN
Use when nothing else works: TB, Septic Peritonitis, Fulminating Pyelonephritis.
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THE WANNABES

All Aminoglycosides end in -mycin...

...but if it has 'THRO' in the middle, THRO it off the list! ⚠

TRUE MYCINS

Mycin

Mycin

Mycin

THRO

The 'THRO' List:
Azithromycin,
Clarithromycin,
Erythromycin.

True Mycins:
Streptomycin,
Cleomycin,
Tobramycin,
Gentamycin,
Vancomycin,
Clindamycin.

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Toxicity Hazard #1: Ototoxicity

Mycin = Mice (Think Ears)

Monitor: Hearing (#1), Balance, Tinnitus.

Damage to Cranial Nerve 8.

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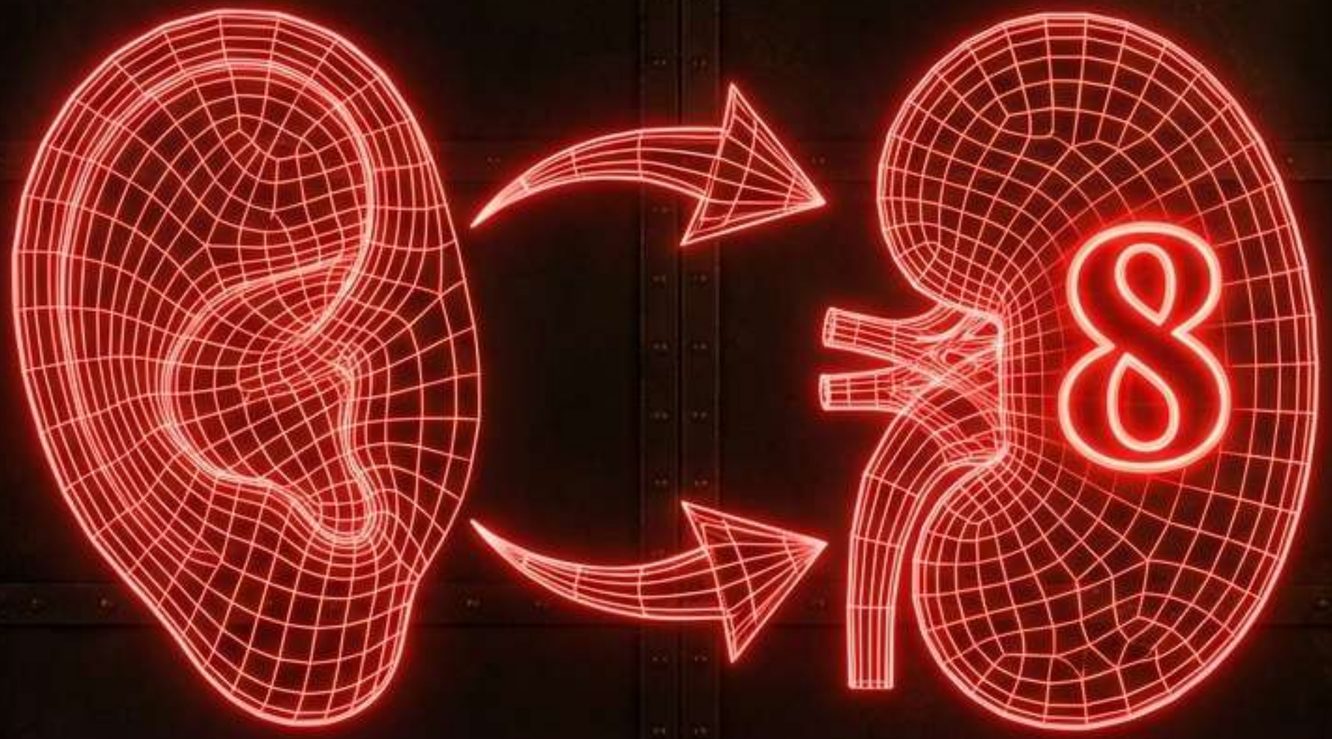
Toxicity Hazard #2: Nephrotoxicity

The human ear is shaped like a kidney.



Monitor: 24-Hour Creatinine Clearance (Best Indicator).

Serum Creatinine (Second Best).

The Number 8: Toxic to CN8, Administer q8h.

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Route of Administration

PO = Not Absorbed = No Systemic Effect.

IM/IV = Systemic Effect (Required for infection).

IM/IV

PO

SYSTEMIC CIRCULATION

Oral (PO) Exceptions:

1. Hepatic Encephalopathy
   (Kill E. Coli -> Lower Ammonia).
2. Pre-op Bowel Surgery.

BOWEL STERILIZATION

Who can sterilize my bowel? Neo-Kan! (Neomycin & Kanamycin).

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THE “TAP” PROTOCOL

Trough -> Administer -> Peak.

Required for Narrow Therapeutic Index (Small difference between healing and killing).

Used for:
Mean Old Mycins & Digoxin.

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Timing the Mechanism

TROUGH: Always 30 mins before next dose.

PEAK:

- Sublingual: 5–10 mins after dissolve.
- IV: 15–30 mins after bag finishes.
- IM: 30–60 mins after injection.
- PO: Not necessary.

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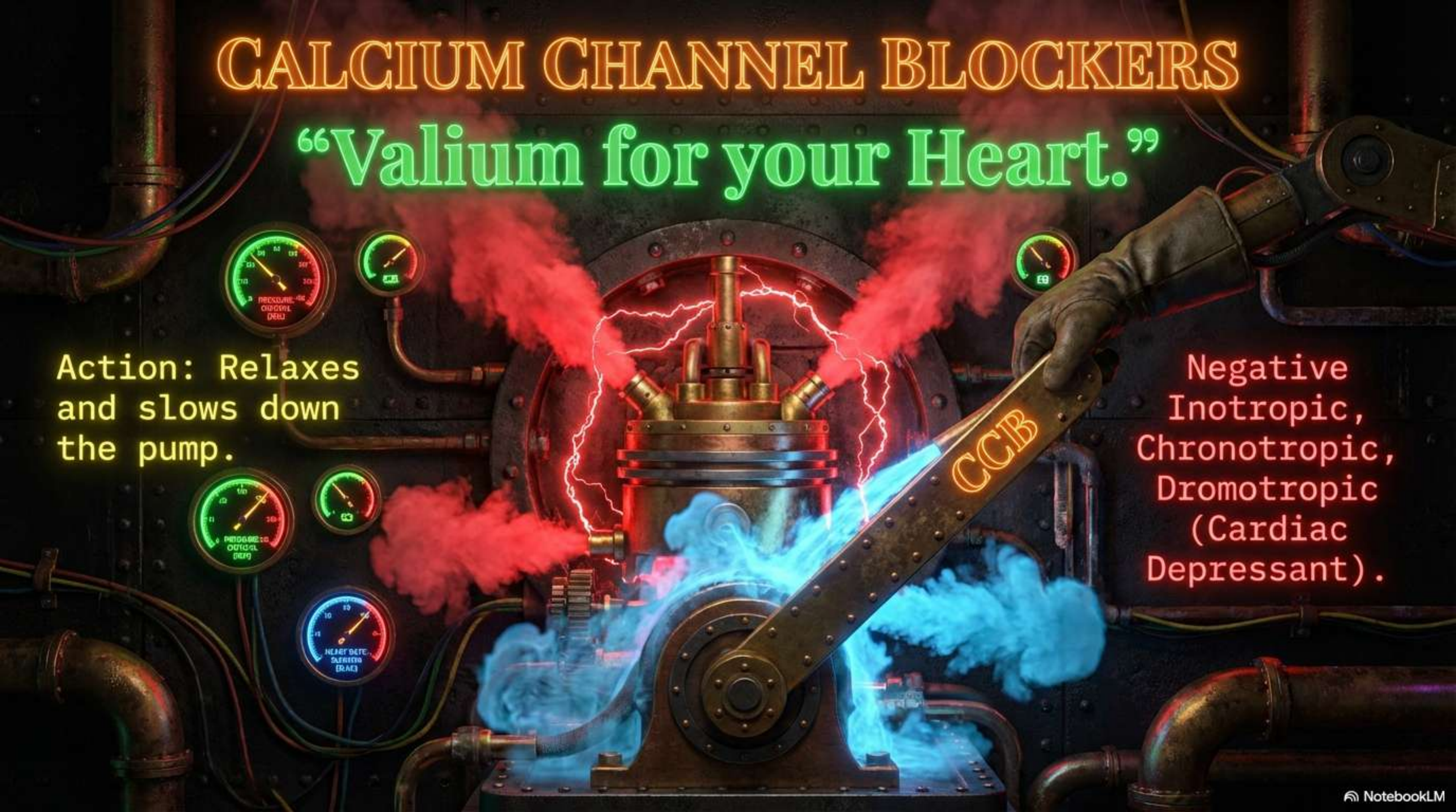
Calcium Channel Blockers

"Valium for your Heart."

Action: Relaxes and slows down the pump.



Negative Inotropic, Chronotropic, Dromotropic (Cardiac Depressant).

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INDICATIONS (What it treats)

Treats Atrial (and SVT).
NOT Ventricular.

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Identification & Safety

Names: End in '-dipine' + Verapamil & Cardizem.

SIDE EFFECTS: H & H (Headache & Hypotension).

CRITICAL NURSING ACTION:

BP: 90/60

Measure Blood Pressure. HOLD if Systolic < 100.
Titrate IV drip to keep SBP > 100.

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The Toxicity VATS: Pattern Recognition

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THE LOW NUMBERS

LITHIUM
(Mania)
| Therapeutic: 0.6 - 1.2
| Toxic: ≥ 2.0

LANOXIN/DIGOXIN
(A-Fib/CHF)
| Therapeutic: 1 - 2
| Toxic: ≥ 2.0

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THE HIGH NUMBERS (10-20)

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Elevated Level: 10 – 20.
Hospitalize at 14–15.
TOXIC: ≥ 20.
BILIRUBIN (Newborns Only)
Jaundice: Yellow skin. Pathological (Birth) vs Physiological (Day 3).

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Critical Failure: Kernicterus

Bilirubin > 20 invades the Brain (CSF).
Aseptic Meningitis.

Sign: OPISTHOTONOS (Hyperextension/Arching).

NURSING ACTION: Place child on their side.

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INSULIN: THE TIMING GEARS

REGULAR (Rapid/Run): Clear solution.
Onset: 1 hr | Peak: 2 hrs | Duration: 4 hrs
Pattern: 1-2-4
Can be IV drip.

NPH (Not so fast): Cloudy suspension.
Onset: 6 hrs | Peak: 8-10 hrs | Duration: 12 hrs
Pattern: 6-8-10-12
Never IV.

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Mixing Protocol

The “RN” Rule: Regular then NPH.

Pressurize N -> Pressurize R -> Draw R -> Draw N

Pressurize N -> Pressurize R -> Draw R -> Draw N

Expiration: 30 days after opening.
Write ‘EXP’ and date.
Refrigeration optional after opening.

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ANTICOAGULANT MATRIX

HEPARIN: IV/SQ. Fast acting.
Antidote: Protamine Sulfate.
Test: PTT.

Safe for Pregnancy

WARFARIN: PO. Slow acting.
Antidote: Vitamin K. Test: PT/INR.

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Survival Summary

1. Respect the therapeutic ranges (Narrow windows kill).
2. Draw TAP on Mycins. Hold CCBs if BP < 100.
3. Watch for Kernicterus in newborns > 20.
4. Know your Antidotes and Peaks.

LAB CLOSED.

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